Frontier VLMs are Strong Zero-Shot Soccer Video Reasoners with Proper Elicitation

Xingyu Zhu[1,2], Yu Zhang[1], Wenwu He[1], Yuyang Sun[3], Haoxuan Ma, Yongliang Wu[3], Xiaogang Wang[4], Xinyu Ye[5], Zhenxiang Jiang, Yangguang Ji, Wenbo Zhu, Xu Yang[3]
[1]Beijing Freedo Technology Co., Ltd. [2]National University of Singapore
[3]Southeast University [4]Southwest University [5]UNC-Chapel Hill

Abstract

We present a task-routed pipeline for the SoccerNet Video Question Answering (VQA) 2026 Challenge, built upon Gemini-3.1-Pro as a unified multimodal reasoning backbone. Rather than training task-specific models or orchestrating multiple specialized agents, we focus on eliciting the latent perception and reasoning capabilities of a single frontier vision-language model (VLM) through carefully designed prompting strategies and lightweight auxiliary tools. We identify four elicitation paradigms that collectively cover all 14 challenge tasks, knowledge grounding, visual prompting, reasoning decomposition, and temporal understanding. Each task is routed to the paradigm best suited to its modality and reasoning demands, with external tools such as YOLOv8x and hybrid database-web retrieval serving as perceptual scaffolds that sharpen the VLM's attention rather than replacing its reasoning. Our approach achieves first place on the challenge leaderboard with an overall accuracy of 97.6% on the 500-question test set.

1. Method

Our system is built on a single design principle that frontier VLMs already possess strong multimodal perception and reasoning capabilities, and the key challenge is to *unlock* these capabilities for each task through appropriate elicitation strategies. As illustrated in Figure 1, our pipeline consists of a task router that dispatches each question to a task-specific strategy, auxiliary tools that provide structured scaffolding to guide the VLM's attention, and Gemini-3.1-Pro [3] as the unified reasoning engine that produces final answers conditioned on the enriched inputs. This paradigm-driven design is inspired by but distinct from multi-agent approaches such as SoccerAgent [1], which employs 18 specialized tools coordinated by a separate orchestrator. We instead consolidate all reasoning within a single VLM, using lightweight external tools only as *perceptual scaffolds* rather than independent decision-makers. We detail our four elicitation paradigms in the following subsections.

1.1. Knowledge Grounding

Several tasks require to reason about factual knowledge that cannot be directly perceived from visual input alone. We address this by injecting structured external knowledge into context, transforming open-ended recall into grounded reasoning over retrieved evidence.

The most representative case is the pure-text QA task, where we design a two-stage retrieve-then-reason pipeline. For match-specific queries, we perform fuzzy matching against a local database of game metadata and event timelines, then concatenate the results with Google Search grounding, allowing the VLM to cross-validate local records against web sources.

A similar principle applies to image-based knowledge QA , where we adopt a reverse reasoning strategy, first retrieving identifying attributes for each answer candidate via web search, then letting the VLM match these profiles against visual evidence. For camera type classification, we inject a complete taxonomy of 13 camera types with disambiguation rules, converting ambiguous perception into structured classification.

1.2. Visual Prompting

When tasks demand fine-grained spatial reasoning in cluttered scenes, the VLM's free-form perception often lacks sufficient precision. We address this by augmenting input images with structured visual annotations that direct the model's attention.

The key innovation lies in player counting . We transform this open-ended counting problem into structured classification by introducing numbered bounding boxes as visual prompts. YOLOv8x [2] detects all persons in the frame, then we render numbered green bounding boxes with non-overlapping leader lines onto the original image. The VLM receives both the annotated image and a structured

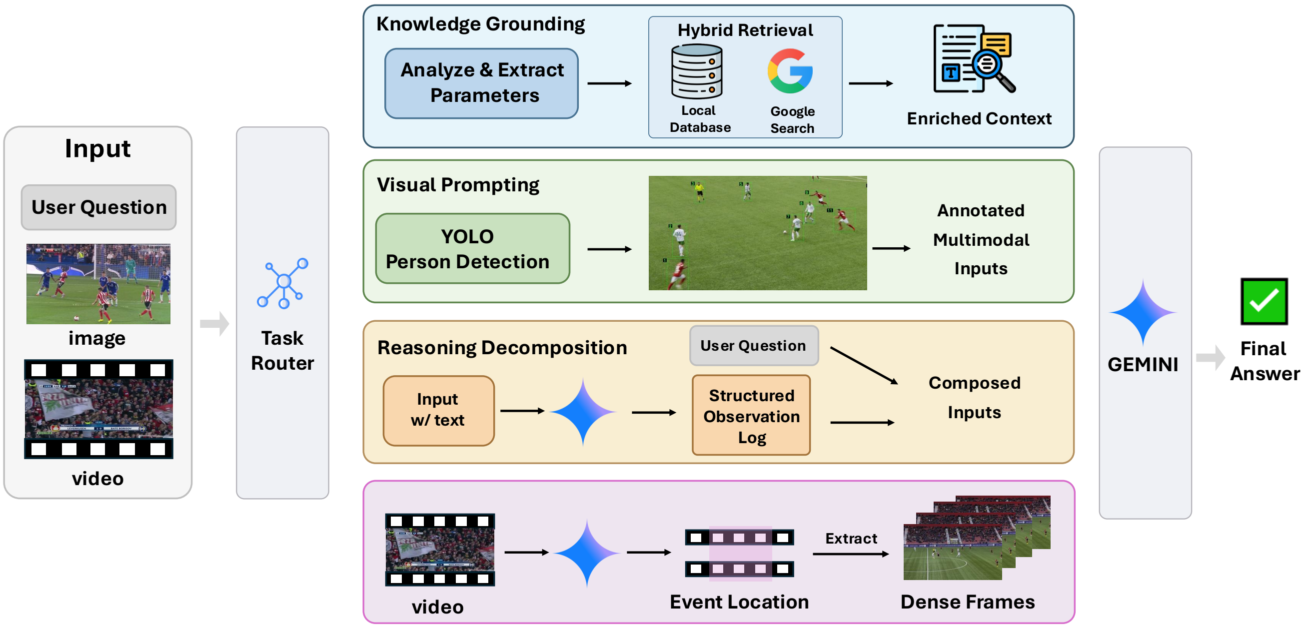

Figure 1. Overview of our task-routed VLM elicitation pipeline. Each question is dispatched to one of four paradigms that provide structured scaffolding to Gemini-3.1-Pro.

person list, classifying each numbered individual by role rather than counting freely.

For jersey number recognition , we upload 8 diverse frames simultaneously so the VLM can cross-reference partial digit observations across viewpoints. Score and time reading uses attention-directing prompts that prioritize broadcast overlay regions.

1.3. Reasoning Decomposition

Some tasks exhibit a subtle failure mode where the VLM's perception becomes biased by the answer options it simultaneously evaluates. We mitigate this through multi-round reasoning that separates observation from judgment.

Jersey color QA requires precise disambiguation of team identity and color naming. We employ two-round reasoning, where the first round localizes the key action segment and identifies the teams involved, then retrieves official jersey color information from our local database as a reference anchor. We re-encode the relevant segment at a reduced frame rate and densely sample 12 frames. The VLM examines these in the second round, combining database-retrieved color context with direct visual observation. Careful prompt design addresses confounding factors such as home and away team confusion and color synonym mapping.

1.4. Temporal Understanding

Video tasks require temporal reasoning, where we identify frame sampling density as the critical bottleneck rather than reasoning capacity. We adopt an adaptive strategy that adjusts sampling granularity based on video duration.

For short clips under 8 seconds, we re-encode at a reduced frame rate to produce slow-motion versions with approximately 8 evenly spaced frames, ensuring sufficient temporal coverage for brief actions. For longer videos, we adopt a coarse-to-fine strategy where the VLM first identifies the key temporal segment, then we densely sample frames from this window for fine-grained analysis.

Multi-view foul recognition extends temporal reasoning across camera angles, with each viewpoint independently analyzed and per-view judgments consolidated through majority vote. Replay grounding uploads all five candidate clips simultaneously, enabling temporal overlap identification in a single pass.

2. Implementation Details

We use Gemini-3.1-Pro Preview with a thinking budget of 10,000 tokens and temperature 0. The local match database is adopted from SoccerAgent [1].

3. Results

Our approach achieves an overall accuracy of 97.6% on the 500-question test set, ranking first on the SoccerNet VQA 2026 Challenge leaderboard.

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